A Proof by Contradiction on the Non-Integer Property of Partial Harmonic Sums

Srivatsan Sriram

November 9, 2026

Abstract

We present a proof by contradiction that the partial sum of the harmonic series, $S_n = \sum_{i=2}^{n} \frac{1}{i}$, is never an integer for any integer $n \geq 2$. The argument compares the sums at two consecutive indices k and $k+1$, splits into two cases according to whether $k+1$ divides $L_k = \mathrm{LCM}(2, 3, \ldots, k)$, and finishes with a parity analysis based on 2-adic valuations.

1 Introduction

The fact that harmonic numbers are never integers for $n > 1$ is a classical result in number theory. A standard proof considers the unique term whose denominator is divisible by the highest power of 2. In this note we give a variant of that 2-adic argument, organised around the step from S_k to S_{k+1}. Assuming that S_{k+1} is an integer, we express S_k as a fraction with denominator $k+1$ in lowest terms and compare this with the fact that the denominator of S_k divides $L_k = \mathrm{LCM}(2, \ldots, k)$. A parity argument based on the identity $L_k S_k = m\,[a(k+1) - 1]$ then yields the contradiction. The proof uses the global quantities L_k and $\alpha = \max_{2 \leq i \leq k} v_2(i)$.

2 Notation and a lemma

For a positive integer x, let $v_2(x)$ denote the largest integer $e \geq 0$ such that $2^e \mid x$. We use three standard facts.

(V1) $v_2(xy) = v_2(x) + v_2(y)$.

(V2) Every positive integer x can be written as $x = 2^{v_2(x)}\, b$ with b odd.

(V3) $v_2\big(\mathrm{LCM}(x_1, \ldots, x_r)\big) = \max_j v_2(x_j)$.

Lemma 1. *Let $k \geq 2$, let $L_k = \mathrm{LCM}(2, 3, \ldots, k)$, and let $\alpha = \max_{2 \leq i \leq k} v_2(i)$. Then*

(a) $\alpha \geq 1$ and $v_2(L_k) = \alpha$;

(b) if $i \in \{2, \ldots, k\}$ and $v_2(i) = \alpha$, then $i = 2^\alpha$; moreover $2^\alpha \in \{2, \ldots, k\}$, so $2^\alpha \leq k$;

(c) $k < 2^{\alpha+1}$.

Proof. (a) The element 2 lies in $\{2, \ldots, k\}$ and $v_2(2) = 1$, so $\alpha \geq 1$. By (V3), $v_2(L_k) = \max_{2 \leq i \leq k} v_2(i) = \alpha$.

(b) Let $i \in \{2, \ldots, k\}$ with $v_2(i) = \alpha$. By (V2), $i = 2^\alpha c$ with c odd. Suppose $c \geq 3$. Then $2 \leq 2^{\alpha+1} = 2 \cdot 2^\alpha < 3 \cdot 2^\alpha \leq i \leq k$, so $2^{\alpha+1} \in \{2, \ldots, k\}$ and $v_2(2^{\alpha+1}) = \alpha + 1 > \alpha$, contradicting the definition of α as a maximum. Hence $c = 1$ and $i = 2^\alpha$. Since the maximum α is attained by at least one element of $\{2, \ldots, k\}$, that element equals 2^α; so $2^\alpha \in \{2, \ldots, k\}$.

(c) If $2^{\alpha+1} \leq k$, then $2^{\alpha+1} \in \{2, \ldots, k\}$ (as $2^{\alpha+1} \geq 2$) and $v_2(2^{\alpha+1}) = \alpha + 1 > \alpha$, again a contradiction. Hence $k < 2^{\alpha+1}$. □

3 Main Theorem

Theorem 1. *Let $S_n = \sum_{i=2}^{n} \frac{1}{i}$. For all integers $n \geq 2$, $S_n \notin \mathbb{Z}$.*

Proof. Suppose for the sake of contradiction that there exists some integer $n \geq 2$ such that $S_n \in \mathbb{Z}$. Let us denote this target index as $n = k + 1$, where $k \geq 1$. By our initial assumption, we can write:

$$S_{k+1} = \sum_{i=2}^{k+1} \frac{1}{i} = a$$

for some integer $a \in \mathbb{Z}$. We isolate the final term of the summation to evaluate the structural relationship between successive states:

$$S_k + \frac{1}{k+1} = a \implies S_k = a - \frac{1}{k+1} = \frac{a(k+1) - 1}{k+1} \tag{1}$$

Notice that the numerator and denominator of the resulting fraction are coprime, since $\gcd(a(k+1) - 1, k+1) = \gcd(-1, k+1) = 1$. Thus the right-hand side of (1) is a fraction in lowest terms, and the denominator of S_k in lowest terms is exactly $k + 1$.

Let $L_k = \mathrm{LCM}(2, 3, \ldots, k)$, with the convention that the empty LCM equals 1, so $L_1 = 1$. We partition our analysis into two mutually exclusive and exhaustive cases based on the divisibility relationship between the successor term $(k + 1)$ and L_k.

Case 1: $(k + 1) \nmid L_k$

Each $i \in \{2, \ldots, k\}$ divides L_k, so $L_k S_k = \sum_{i=2}^{k} \frac{L_k}{i}$ is an integer. Therefore the denominator of S_k in lowest terms divides L_k. By Equation (1) that denominator is exactly $k+1$, so $(k+1) \mid L_k$. This contradicts the case assumption $(k + 1) \nmid L_k$. Hence Case 1 is impossible.

In particular, for $k = 1$ we have $L_1 = 1$ and $k + 1 = 2 \nmid 1$, so $k = 1$ falls under Case 1 and is already excluded. (Directly: $S_2 = \frac{1}{2}$.) Consequently, in Case 2 below we may and do assume $k \geq 2$.

Case 2: $(k + 1) \mid L_k$ (so $k \geq 2$)

If $(k + 1)$ divides L_k, there exists an integer multiplier $m \geq 1$ such that:

$$L_k = m(k + 1) \tag{2}$$

Multiplying both sides of the structural identity in Equation (1) by L_k, we obtain:

$$L_k \cdot S_k = m(k+1) \cdot a - m \implies L_k \cdot S_k = m[a(k+1) - 1] \tag{3}$$

Let $\Delta = L_k \cdot S_k = \sum_{i=2}^{k} \frac{L_k}{i}$, an integer. We analyze the parity of both sides of Equation (3).

Let $\alpha = \max_{2 \leq i \leq k} v_2(i)$. By Lemma 1, $v_2(L_k) = \alpha$, and within $\{2, 3, \ldots, k\}$ the element 2^α is the unique element attaining the maximum valuation. Expanding the summation for Δ:

$$\Delta = \frac{L_k}{2} + \frac{L_k}{3} + \cdots + \frac{L_k}{2^\alpha} + \cdots + \frac{L_k}{k}$$

By (V1), $v_2\left(\frac{L_k}{i}\right) = \alpha - v_2(i)$ for each $i \in \{2, \ldots, k\}$. For every term with $i \neq 2^\alpha$ we have $v_2(i) < \alpha$ (since $v_2(i) \leq \alpha$ by maximality, and equality would force $i = 2^\alpha$ by Lemma 1(b)), so $v_2\left(\frac{L_k}{i}\right) \geq 1$ and the quotient $\frac{L_k}{i}$ is an *even* integer. For the unique term $i = 2^\alpha$, we have $v_2\left(\frac{L_k}{2^\alpha}\right) = 0$, so the quotient is an *odd* integer. The sum of any number of even integers and exactly one odd integer is **odd**. Therefore, Δ is odd.

We now evaluate the parity of the right-hand side of Equation (3), $m[a(k+1)-1]$, by determining the properties of the multiplier m. By (V1) applied to Equation (2),

$$\alpha = v_2(L_k) = v_2(m) + v_2(k+1).$$

- **Case 2(a): $(k+1)$ is an odd integer.**
  Then $v_2(k+1) = 0$, so $v_2(m) = \alpha \geq 1$ by Lemma 1(a). (Equivalently: L_k is even because $2 \in \{2, \ldots, k\}$.) Thus, m is an **even** integer.

- **Case 2(b): $(k+1)$ is an even integer.**
  Write $k+1 = 2^\beta \cdot b$ with $\beta = v_2(k+1) \geq 1$ and b odd, by (V2). Since $(k+1) \mid L_k$, we have $\beta \leq v_2(L_k) = \alpha$.

  First, $(k+1)$ is not a pure power of 2, i.e. $b \neq 1$. Indeed, if $b = 1$ then $k+1 = 2^\beta$ with $\beta \leq \alpha$, so $k+1 \leq 2^\alpha \leq k$ by Lemma 1(b), which is absurd. Hence $b > 1$, and since b is odd, $b \geq 3$.

  Next, $\beta < \alpha$. We already know $\beta \leq \alpha$. Suppose $\beta = \alpha$. Then $k+1 = 2^\alpha b \geq 3 \cdot 2^\alpha$, so $k \geq 3 \cdot 2^\alpha - 1 \geq 2 \cdot 2^\alpha = 2^{\alpha+1}$, contradicting Lemma 1(c). Hence $\beta < \alpha$.

  From $\alpha = v_2(m) + \beta$ and $\beta < \alpha$ we get $v_2(m) = \alpha - \beta \geq 1$. Consequently, the multiplier m is an **even** integer.

Since m is even in both subcases, the entire right-hand side product $m[a(k+1)-1]$ is an **even** integer. This establishes a contradiction, as the odd left-hand side Δ cannot equal an even integer.

Conclusion

Both cases lead to a contradiction, so our initial assumption that there exists an index $n \geq 2$ with $S_n \in \mathbb{Z}$ must be false. Therefore, $S_n \notin \mathbb{Z}$ for all integers $n \geq 2$. □

Formal verification. The proof has been formalized and machine-checked in Lean 4 with Mathlib; the source is available at `https://github.com/srivatssriram/harmonic-non-integer`.